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Research article

An explainable feature selection framework for web phishing detection with machine learning

Sakib Shahriar Shafin

Centre for Smart Analytics, Federation University Australia, Ballarat, VIC 3356, Australia

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ABSTRACT

In the evolving landscape of cyber threats, phishing attacks pose significant challenges, particularly through deceptive webpages designed to extract sensitive information under the guise of legitimacy. Conventional and machine learning (ML)-based detection systems struggle to detect phishing websites owing to their constantly changing tactics. Furthermore, newer phishing websites exhibit subtle and expertly concealed indicators that are not readily detectable. Hence, effective detection depends on identifying the most critical features. Traditional feature selection (FS) methods often struggle to enhance ML model performance and instead decrease it. To combat these issues, we propose an innovative method using explainable AI (XAI) to enhance FS in ML models and improve the identification of phishing websites. Specifically, we employ SHapley Additive exPlanations (SHAP) for global perspective and aggregated local interpretable model-agnostic explanations (LIME) to determine specific localized patterns. The proposed SHAP and LIME-aggregated FS (SLA-FS) framework pinpoints the most informative features, enabling more precise, swift, and adaptable phishing detection. Applying this approach to an up-to-date web phishing dataset, we evaluate the performance of three ML models before and after FS to assess their effectiveness. Our findings reveal that random forest (RF), with an accuracy of 97.41% and XGBoost (XGB) at 97.21% significantly benefit from the SLA-FS framework, while *k*-nearest neighbors lags. Our framework increases the accuracy of RF and XGB by 0.65% and 0.41%, respectively, outperforming traditional filter or wrapper methods and any prior methods evaluated on this dataset, showcasing its potential.

1. Introduction

With increasing dependence on online services for various aspects of our lives, the frequency and complexity of cyber threats targeting personal data are rising. Phishing is a common and significant threat where attackers deceive individuals into revealing sensitive information such as passwords and bank details (Khonji et al., 2013). This type of cybercrime has become more advanced, with attackers creating fake emails and websites that appear legitimate. Many webpage-based phishing attacks now mimic security and certificates like genuine websites, making it harder to detect harmful online environments. The spread has been so severe that about 4.7 million phishing attack attempts were logged in 2022 alone (Smith, 2024), and this number is constantly increasing. Awareness and understanding of how to protect against threats are still lacking among the public, making phishing one of the most effective and damaging attack types today.

In the face of increasingly sophisticated phishing attacks, traditional mechanisms such as blacklists and heuristic rules are becoming somewhat ineffective (Deval et al., 2021). This imbalance underscores the pressing need for advanced solutions such as machine learning (ML). ML stands out by effectively identifying phishing sites' anomalous patterns and deceptive structures. By building ML models trained on a host of phishing data and associated features, existing and evolving phishing attacks can be detected more accurately (Do et al., 2022). However, the biggest challenge lies in effectively choosing specific characteristics for a detection model to focus on. A successful detection system should be able to extract necessary and critical information for the ML model to work on. Otherwise, faulty detection will allow the theft of important sensitive information or trigger false alerts, hampering legitimate online activities. As such, identifying the most relevant and impactful features for the ML algorithms is a significant challenge.

The implementation of ML and deep learning algorithms has become a common approach for web phishing detection. Various feature selection (FS) methods have been conceptualized and evaluated to enhance the performance of these models by focusing on the most valuable traits toward classification. FS methods can be broadly classified into three categories: (1) filter methods, which employ tests like chi-square and mutual information to evaluate feature relevance by measuring statistical

E-mail address: ss.shafin@federation.edu.au.

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significance (Lyu et al., 2023); (2) wrapper methods, which select multiple feature subsets and apply model performance metrics such as accuracy or F1-score to ultimately select the best-performing set (Maldonado et al., 2022); and (3) metaheuristic optimization methods, which use algorithms like genetic algorithms or particle swarm optimization to iteratively refine feature sets by minimizing or maximizing a defined objective function (Hu et al., 2023). Many of these types have been employed in web phishing detection research.

However, traditional FS methods often fail to attain actual improvement and decrease model performance when the attack types are complex. This is an issue for webpage phishing detection, as information gathered to detect whether a website is phishing consists of multivariate information like URL, website content, and external indexing features. Thus, reducing critical information when detecting an already complex and evolving type of attack can result in misclassification by the ML models. This can be seen in Shabudin et al. (2020) and Moedjahedy et al. (2022), where the classification accuracy decreases by applying filter-based FS on web phishing datasets. In this case, while an argument can be made that the reduced number of features helps identify attacks quicker (as evidenced by the increased detection speed), the main goal, which is to enhance model accuracy, is unrealized. Moreover, traditional FS methods typically lack the flexibility to adapt to local, context-specific variations in feature importance. Wrapper and metaheuristic methods, while effective in ranking features, do not consider the individual context of samples, limiting their adaptability to changing attack patterns (Kuzudisli et al., 2023). In real-world network scenarios, the impact of different phishing attacks varies. System operators need the flexibility to focus on specific attack types and select features relevant to those classes, a capability that most existing FS models for webpage phishing detection lack. This gap calls for an FS method that can enhance attack detection accuracy, contain a local view of data, and be tuned to perform class-specific FS.

The emergence of explainable AI (XAI) introduces a shift toward interpretable ML development, among which SHapley Additive exPlanations (SHAP) is a popular method (Dikshit and Pradhan, 2021). Based on game theory, SHAP quantifies the contribution of each feature in a dataset to a model's prediction, offering a granular understanding of their individual effects. Each feature's contribution to each class can be visualized, helping system designers identify the impactful ones. On the other hand, the local interpretable model-agnostic explanations (LIME) technique provides local explanations by approximating the model locally around a prediction. It generates interpretable models for individual predictions, allowing an understanding of feature importance on a case-by-case basis, which is crucial for tracking changing attack patterns. Despite the unique merits of these models, explainable FS for webpage phishing detection is rarely employed.

Recognizing their combined global and local explainability and class-wise adaptability, we propose SHAP and LIME-aggregated FS (SLA-FS), a novel FS framework that combines SHAP's global interpretability with an aggregated version of LIME's local analysis to create an innovative FS method for webpage phishing detection with class-specific feature importance. Initially, SHAP ranks features based on their contribution to detecting phishing attacks. After obtaining the SHAP ranking, LIME validates the importance of these features from a local perspective to ensure there is no bias in the selection from other classes. The final feature set is then selected by cross-checking the top features identified by both methods. This framework is evaluated using the recent web page phishing dataset. We compare its performance with traditional and pre-existing dataset-specific FS methods in an extensive evaluation using three prominent ML algorithms: random forest (RF), XGBoost (XGB) and *k*-nearest neighbors (KNN). To the best of our knowledge, this is the first study to develop an FS framework integrating SHAP and LIME and comprehensively evaluate its effectiveness in web phishing detection.

The contributions of this study can be summarized as follows:

- This study proposes an innovative and interpretable FS method integrating the SHAP and LIME algorithms to extract global and local class-specific feature importance for enhanced webpage phishing attack detection.
- A detailed comparative assessment with existing FS methods is performed by employing three ML models (RF, XGB, and KNN) on each FS-derived feature set using the web page phishing dataset.
- Result analysis with both traditional FS methods and previously conceptualized frameworks on the dataset demonstrates that our model, combined with RF, attains the highest detection accuracy (97.41%) and the highest detection improvement post-FS (0.65%).

The remainder of the paper is organized as follows. Section 2 presents the related works on ML applications and FSs for phishing attack detection. Section 3 details the study methodology, while Section 4 contains the evaluation results and analysis of the proposed method. Finally, Section 5 summarizes the conclusion and indicates future studies.

2. Related works

In cybersecurity, phishing, particularly webpage phishing, is a significant and evolving threat that targets individuals directly. Its constant evolution complicates distinguishing between legitimate and malicious sites. ML models have emerged as superior to traditional detection systems in identifying phishing attempts, attaining better detection performance (Patil et al., 2018).

FS has been identified as a critical factor in improving the performance of ML models for detecting and classifying cyberattacks (Shafin et al., 2021b). A detailed analysis by Sharma et al. (2020) assessed ML models, including RF, KNN, and support vector machine, with and without FS techniques on web phishing datasets. Despite using the chi-square test, recursive feature elimination (RFE), and other FS methods across two publicly available datasets, the outcomes revealed that FS did not consistently enhance, and sometimes even reduced the performance of ML models. To find an improved FS method in the domain of web phishing detection, Moedjahedy et al. (2022) introduced a two-stage FS method, placing special emphasis on URL-related features. Initially, this method eliminates less important features by removing those with high correlation. Subsequent application of RFE refines the feature set further. Tested on publicly available datasets, the method ultimately selected 10 key features for ML analysis. Despite a reduction in prediction time, accuracy declined, for example, from 98.1% to 96% on dataset 1, with similar trends observed in dataset 2, indicating a decrease in model performance.

A hybrid FS framework was outlined in Bhowmik and Bhowmik (2022), combining four filter methods in the initial phase and integrating wrapper and embedded methods in the subsequent stage. This approach employs a set union operation for aggregating initial features and an intersection operation to select final features for ML algorithms. Although direct comparisons pre- and post-FS were not presented, this model does surpass existing studies in accuracy. However, its extensive FS process is time-intensive, and the complex interplay between an array of FS methods risks overfitting and feature correlation, potentially omitting critical detection information.

Using a contemporary dataset for attack detection cannot be overstated. Hannousse and Yahiouche (2021a) introduced a benchmark dataset, compiled through web crawlers and enhanced with document object model analysis to address the transient nature of phishing websites. The collected dataset underwent relevance and runtime analysis to ensure research applicability. Experiments show RF's superior performance (96.83% accuracy) when used alongside filter-based feature ranking and elimination techniques, rather than wrapper methods. Due to the comprehensive set of features, such as URL-based, content-based, and external-based, we use this dataset as the base for evaluation. A preliminary study on the dataset was conducted in Alrefaai et al. (2022), where a diverse number of ML models including decision tree, RF, gradient boosting (GB), KNN, XGB, and multilayer perceptron were employed. The highest detection accuracy achieved was, however, 96.60% by XGB, placing it under the results attained by the original study.

To enhance ML performance on the dataset, Adane et al. (2023) applied univariate feature selection (UFS) to evaluate the efficacy of ensemble classifiers, including RF, GB, and CatBoost. Chosen for its simplicity and speed, UFS, a filter-based FS method, allows rapid evaluation by analyzing individual features' statistical significance without the computational complexity of multivariate approaches. When deploying RF on the dataset, an accuracy rate of 96.54% was reported post-UFS application, which is still below the highest accuracy obtained in Hannousse and Yahiouche (2021a). Nonetheless, among the tested classifiers, RF achieved the fastest response time.

The above discussion highlights a clear gap in traditional FS methods' ability to ML models for detecting phishing websites, pointing to the need for new strategies. Here, XAI comes into play, able to pinpoint important features for phishing detection, as shown in Wei and Sekiya (2022). The researchers aimed to build a more accurate detection framework by combining usual feature importance techniques like mean decrease in impurity and permutation with SHAP for a feature ranking, helping to decide the optimal number of features needed to achieve the best accuracy. Using the Mendeley dataset and RF to test FS models, accuracy improved from 96.17% with 48 features to 97.78% by selecting only 23 features.

Although existing literature on XAI utilization for web phishing detection is considerably scarce, researchers have adopted interpretable FS to strengthen detection for generalized attack detection, specifically in the Internet of Things (IoT). Wang et al. (2020) employed SHAP to gather explanations of the NSL-KDD dataset features. The authors constructed two neural network models and produced predictions, which were then interpreted to extract feature contributions using SHAP. Results demonstrate that the insights derived by SHAP are consistent with the specific attack types, validating its usage. Similarly, Bahadoripour et al. (2024) used SHAP for FS in a federated ML framework designed for industrial IoT networks, where the proposed model achieved a 4.9% accuracy increase due to SHAP.

On the other hand, LIME has been primarily used for localized explanations, as demonstrated in Bora et al. (2024). Recently, however, it has increasingly been applied for global FS processes with many data samples. In Shin (2023), a points system was utilized for LIME, where each sample was evaluated to award points to significant features, culminating in an overall tally to create a ranking. The results indicated that LIME-utilized models outperformed others using wrapper and filter FS methods in accuracy. In Akintade et al. (2023), both LIME and SHAP were employed for attack detection, but not specifically for FS. Instead, they were used to validate insights derived from the primary permutation-based FS model. Additionally, interpretable feature engineering without explicit XAI models has been explored in Wu et al. (2024). This study aimed to identify the linear relationships between highly volatile elements of wind speed and environmental components using the Pearson correlation coefficient. At the same time, RFE was applied to capture nonlinear relationships.

The adoption of XAI-powered FS has demonstrated a gradual shift in ML-driven phishing detection. The shortcomings of traditional methods necessitate innovative strategies, with XAI offering interpretability, preciseness, and tunability for system operators or designers, making it a suitable candidate for wide applicability. While the mentioned studies provide innovative solutions, a comprehensive, explainable, and class-specific FS method for webpage phishing detection seems to not exist in current literature.

3. Research methodology

This section details the SLA-FS framework, designed to enhance FS for phishing attack detection by combining SHAP and LIME algorithms. We also provide an overview of the subsequent ML models used to evaluate the impact of SLA-FS, ensuring a comprehensive assessment method. The process is conducted in two stages: (1) FS using SLA-FS and (2) evaluating ML models on the reduced dataset to compare performance. An overview of the mechanism is illustrated in Fig. 1.

3.1. SLA-FS framework

Due to extensive online activities, web phishing detection systems frequently process vast amounts of data. The dynamic nature of these websites thus challenges the decision-making processes in identifying phishing attempts. Hence, focusing on the most critical information to detect malicious sites is crucial for the models. In this study, we utilize the specific abilities of SHAP and LIME algorithms to identify the valuable information.

3.1.1. Feature importance derivation with SHAP

SHAP derives a global view of feature importance based on a game theoretic approach (Dikshit and Pradhan, 2021). The Shapley values quantify the contribution of each feature to the model's predictions, offering an overall perspective on which features are most impactful across the entire dataset. SHAP excels in providing class-specific importance, allowing models to focus on specific targets if needed. Additionally, SHAP's model-agnostic characteristic ensures its compatibility across various ML models, making it a good choice for the diverse domain of web phishing detection.

For a given instance, the SHAP value of a feature is the average marginal contribution of this feature across all possible combinations (Roshan and Zafar, 2021). Formally, the SHAP value for feature i in prediction j can be represented as

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} \left[f_{S \cup \{i\}}(x_{S \cup \{i\}}) - f_S(x_S) \right] \tag{1}$$

where F is the set of all features, S is a subset of features excluding i, $f_{S \cup \{i\}}(x_{S \cup \{i\}})$ is the model prediction with feature i included, and $f_S\ (x_S)$ is the prediction without feature i.

Drawing from the findings of Sharma et al. (2020) and Adane et al. (2023), we use the rapid and efficient RF classifier to produce the predictions for SHAP. First, a library version of RF from the Sklearn library is trained on the dataset, after which the SHAP model uses the class predictions to calculate the SHAP values for the importance of each feature. SHAP provides a detailed and actionable analysis of how features influence both benign and malicious class instances, enabling targeted FS based on class-specific effects if required. Here is when the tunability aspect kicks in. In a dynamic network environment, if some feature contributions to phishing detection changes, they can be easily visualized, as shown in Fig. 2 (for the evaluating dataset). Based on these changes, system designers can modify the focus of the detection model.

3.1.2. Feature validation with LIME

While SHAP effectively captures a global perspective of feature importance, it often overlooks local variations and individual prediction contexts. Given phishing websites' diverse appearance and behavior, an FS method must adapt to global patterns and local nuances. To achieve this, we incorporate LIME as the second stage of the SLA-FS framework. LIME provides local interpretability by approximating the model around each prediction, highlighting the importance of features for individual instances. These local insights are aggregated and compared with SHAP's global findings to validate the robustness of the selected features.

LIME approximates black-box ML models locally by generating perturbed data around every individual webpage information x and fitting an interpretable model to these samples. The importance w_i of a feature i in the local surrogate model is defined as

$$w_i = \underset{g \in G}{\operatorname{argmin}} \sum_{(x', y')} \pi(x, x')[f(x') - g(x')]^2 + \Omega(g) \tag{2}$$

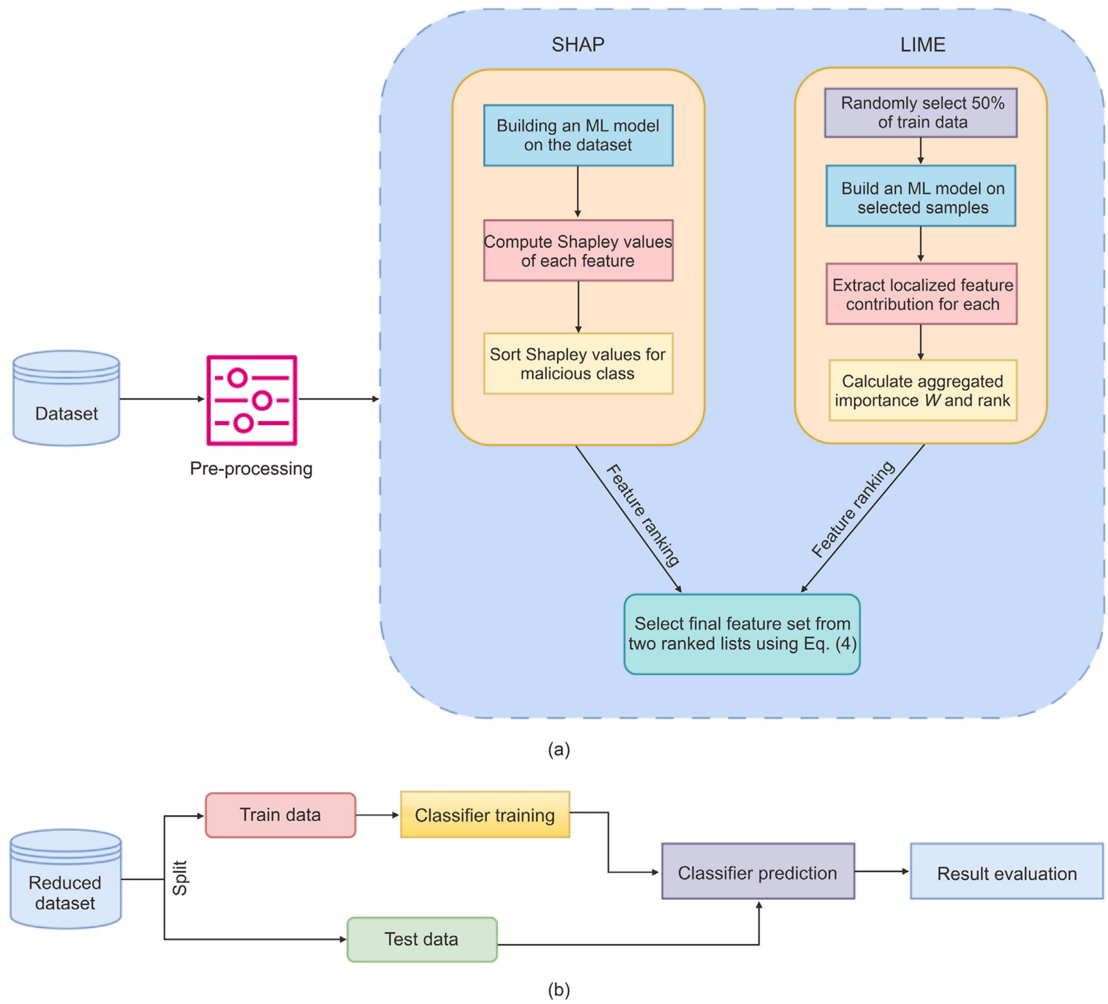

Fig. 1. Overview of the proposed framework. (a) Stage 1: SHapley Additive exPlanations (SHAP) and local interpretable model-agnostic explanations (LIME)-aggreated feature selection; (b) stage 2: model evaluation of dataset with selected features. ML: machine learning.

where G is the class of interpretable models, $\pi(x, x')$ is a proximity measure between the original instance x and the perturbed instance x', $f(x')$ is the black-box model's prediction for x', $g(x')$ is the interpretable model's prediction for x', and $\Omega(g)$ is a complexity penalty for the interpretable model. By minimizing this weighted sum of squared errors, LIME identifies feature weights w_i that explain the local behavior of the black-box model. In the end, the contribution of each i feature value of sample x toward deciding which target class it belongs to is displayed. For implementation in our study to evaluate the readings, we use the same RF classifier (without any parameter tuning), which is also used to produce SHAP feature rankings (as detailed in Section 3.1.1).

However, as LIME only concerns one sample at a time, we need to aggregate the feature weights for all the samples to get collective results, which is necessary for validating SHAP-derived features. To aggregate the insights from LIME across the entire dataset, we first generate local explanations for a subset of observations. Specifically, we randomly sample 50% of the dataset. This fragmented data is denoted as X_f. LIME generates a local surrogate model for each observation $x \in X_f$, yielding feature importance weights w_i^x for each feature i. These weights are aggregated across all sampled observations to determine the overall local importance of each feature.

The aggregated importance W_i for feature i is computed as

$$W_i = \frac{1}{|X_f|} \sum_{x \in X_f} w_i^x \tag{3}$$

The aggregated scores W_i are compared with the SHAP-derived score to validate and finalize selected features.

3.1.3. Final FS

In the final stage, the two resultant score sets (ϕ_i and w_i) are compared to ensure global and local importance. Different systems can set different qualifying thresholds or the number of most-contributing features m to the detection model. The features above this cut-off number are stored in two sets, P_H^m and P_L^m for SHAP and LIME, respectively. A feature i is included in P_H^m if its SHAP score ϕ_i places it in the top m percentile among all features. A similar process is performed for LIME as well. Finally, we define the F_{final} criterion:

$$F_{final} = P_H^m \cap P_L^m \tag{4}$$

This ensures that only the features consistently deemed important by both global (SHAP) and local (LIME) perspectives are retained and passed to the detection models.

We performed experiments to determine the optimal m number of features to keep (Section 4.4.2) and created the resultant dataset. It was then evaluated to identify the most suitable ML algorithm for our FS model. For a detailed comparative analysis, we evaluated three additional FS methods:

- Information gain: This metric evaluates the reduction in entropy or uncertainty about the target variable after observing the feature set,

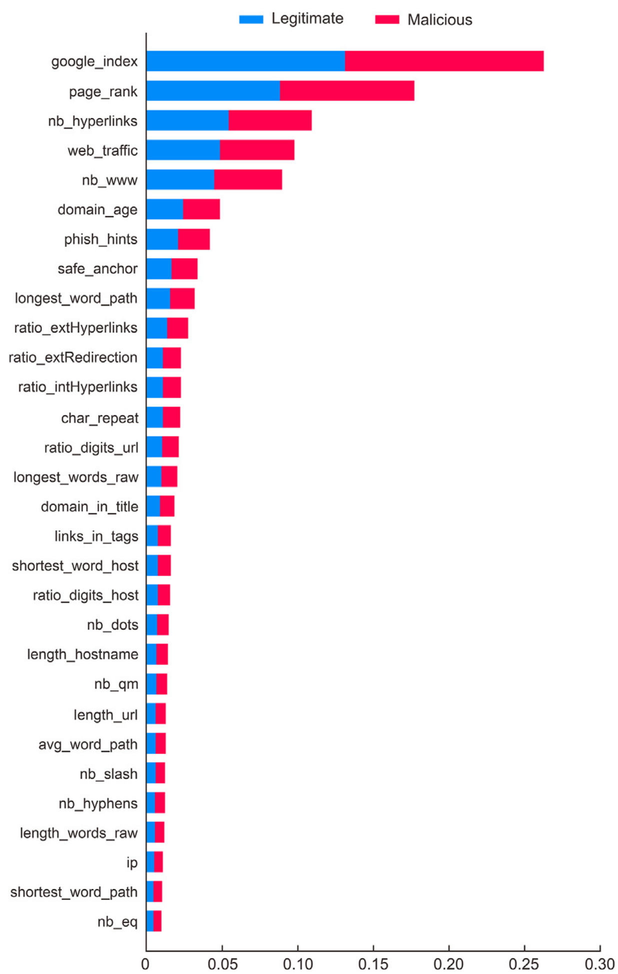

Fig. 2. Top 30 features of web page phishing dataset using SHapley Additive exPlanations (SHAP).

helping to identify features contributing significantly toward class distinction (Yin et al., 2023). The mutual information classifier from Sklearn was used to derive the information gain rankings.

- Chi-square test: This test assesses the independence between categorical variables. We can measure the association of each feature with the target class by monitoring their scores. The *chi*^{2} model was employed under the SelectKBest library from Sklearn to compute the top *m* features using the chi-square test.
- RFE: RFE systematically removes features, building an underlying model with the remaining attributes to identify variables contributing most to prediction (Yin et al., 2023). A default RF classifier was employed, with the standard RFE model and step = 1, denoting one feature removed per iteration.

3.2. Evaluation models

In this study, we compared the performances of three different ML algorithms before and after FS to determine the best one for our web phishing detection framework.

3.2.1. RF

RF is an ensemble learning method that uses the bagging technique (Shafin et al., 2021a), creating a forest of decision trees during training and outputting the class that is the mode of the classes of the individual trees. Bagging brings randomness in building individual trees to ensure model diversity, involving random selection for splitting nodes and reducing overfitting in large feature spaces. This mechanism also makes RF one of the fastest and most resource-efficient ensemble methods. The prediction of RF can be represented as

$$y_e(x) = \frac{1}{N}\sum_{i=1}^{N} y_{t_i}(x) \tag{5}$$

where N is the number of trees, $y_{t_i}(x)$ is the prediction of the i_{th} tree, x is the input feature vector and $y_e(x)$ is the ensemble prediction for that specific vector.

3.2.2. XGB

XGB is an optimized distributed GB library designed to be highly efficient, flexible, and portable. This enhanced boosting technique improves scalability by not requiring linear features or interactions between features and supporting distributed systems (Chowdhury et al., 2023). Furthermore, XGB introduces a more regularized model formalization to control overfitting, strengthening its effectiveness in classification and regression tasks.

3.2.3. KNN's classifier

KNN algorithm is an instance-based learning method that assigns a data point to the most frequently occurring class among its k nearest neighbors in the feature space. This method emphasizes local approximation of the target function, deferring all computations to the classification phase. The efficacy of KNN hinges on two factors: the selection of k and the scaling of features to ensure equitable influence on the classification outcome. A larger k value, while potentially enhancing the robustness to noise, can obscure the decision boundaries between classes, necessitating a careful balance. Another factor is feature scaling, which helps to prevent disproportionate impact from features on a larger scale on the distance computations.

4. Evaluation results

In this section, we present the evaluation results and the derived insights. First, a detailed dataset description is provided, followed by an analysis of the FS process and the subsequent ML performances.

4.1. Dataset

Because of the ever-changing landscape of phishing webpages, an updated and recent dataset is required to assess any detection mechanism properly. In this research, the web page phishing dataset by Hannousse and Yahiouche (2021a) was utilized. It is publicly available in Mendeley (Hannousse and Yahiouche, 2021b) and has garnered considerable attention from researchers. The dataset collects benign samples from Alexa and Yandex while accumulating phishing webpages and associated document object model data from Phishtank and Openphish. The dataset contains a total of 11,430 samples, each with 87 features, of which 56 are URL-based features, 24 are from HTML content extractions, and the remaining 7 are external features obtained from third-party services. The dataset contains two target classes, namely legitimate and phishing.

4.2. Evaluation metrics

Several popular performance metrics used typically for ML (Chowdhury et al., 2021) are evaluated to assess the algorithmm's effectiveness in identifying malicious phishing websites and to measure the ML models' accuracy. The four components of the confusion matrix used in this study are defined as follows:

- True positive (TP): The model accurately recognizes a legitimate website.
- True negative (TN): The model correctly identifies a webpage as malicious.

- False positive (FP): An error occurs when a malicious website is wrongly classified as legitimate.
- False negative (FN): An error occurs when a legitimate website is incorrectly classified as malicious.

Accuracy (ACC): This metric represents the proportion of correctly predicted website classifications (malicious and legitimate) out of the total dataset.

$$ACC = \frac{TP + TN}{N} \times 100 \tag{6}$$

Precision (P): This measures the accuracy of positive predictions.

$$P = \frac{TP}{TP + FP} \times 100 \tag{7}$$

Recall (R): This metric calculates the fraction of actual positive class samples correctly identified by the model.

$$R = \frac{TP}{TP + FN} \times 100 \tag{8}$$

F1-score: A harmonic mean of precision and recall, offering a balance between them for evaluating the model's overall performance.

$$F1\text{-}score = \frac{2 \times P \times R}{P + R} \tag{9}$$

One of the main goals of this study was to increase the efficiency of the models for phishing detection. To that end, we measured the processing time taken by each model to find out the effect of feature set reduction on detection.

4.3. Feature analysis using SLA-FS

Starting with key pre-processing steps, including the conversion of data into numerical format and the removal of null values, we proceeded to feature evaluation.

Taking inspiration from previous studies such as Chiew et al. (2019), where RF was employed to enhanced performance in web phishing detection, we first trained and built and RF classifier on the dataset. The model generates class predictions for each sample, which are then evaluated with SHAP to produce the Shapley values. Fig. 2 visualizes the top 30 features that contribute the most to detecting legitimate (in blue) and malicious websites (in red).

Our study focuses on detecting malicious websites and does not take just a simple anomaly detection approach. Using SHAP's ability to focus on specific classes, we emphasize the features contributing to malicious class detection, not benign ones. Table 1 lists the top 10 features and their SHAP values, highlighting their significance in detecting malicious content. Among these features, four originate from external monitoring, three from website content, and three from URL analysis. This diversity underscores the inadequacy of relying solely on a single data type for web phishing detection, recommending a shift from strictly URL monitoring systems.

The next step is to derive the local-aggregated scores from LIME for the dataset. From the 11,430 samples, 5,715 were chosen randomly and were experimented with LIME. Fig. 3 shows the output for a single random sample.

We aggregate the scores for all the chosen samples, and the top 10 features are shown in Table 2. Some differences from Table 1 are distinctly visible, as phish_hints and ip, ranked 7th and 20th in SHAP evaluation, stands 2nd and 8th in the LIME-aggregated ranking, respectively. This proves that while a global view can produce important insights, localized contributions are also important, which can often be overlooked.

The next step of the evaluation is to select the set of reduced features and create the dataset to be fed into the evaluating ML models to find the most optimal one.

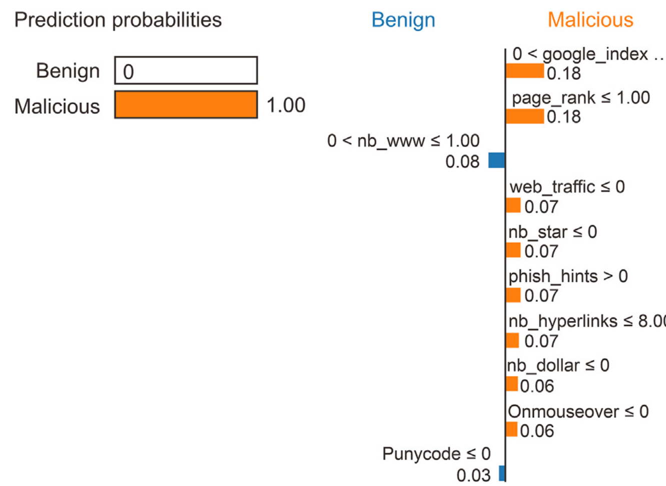

Fig. 3. Local interpretable model-agnostic explanation of sample No. 155 of web page phishing dataset.

Table 1
SHapley Additive exPlanations (SHAP) values of top features.

Feature	Family	SHAP value	Feature description
google_index	External	0.1315	Indexing status in Google
page_rank	External	0.0887	Page rank in Internet databases
nb_hyperlinks	Content	0.0549	Number of hyperlinks in webpage
web_traffic	External	0.0491	Traffic or visitors of the URL
nb_www	URL	0.0452	Multiple usage of "www" in URL
domain_age	External	0.0246	Domain age of the webpage
phish_hints	URL	0.0212	Phish-sensitive words in URL
safe_anchor	Content	0.0173	Anchor HTML linking to javascript
longest_word_path	URL	0.0162	Longest string of words in URL
ratio_extHyperlinks	Content	0.0141	External domain hyperlink ratio

Table 2
Local interpretable model-agnostic explanations (LIME)-aggregated values of top features.

Feature	LIME-aggregated value
google_index	0.0909
phish_hints	0.0509
nb_www	0.0378
page_rank	0.0346
domain_in_brand	0.0247
web_traffic	0.0219
ratio_digits_host	0.0219
ip	0.0153
suspecious_tld	0.0143
longest_word_path	0.0126

4.4. Detection results

This section presents the results from our experiments, which assess the effectiveness of using XAI (SHAP) for FS in webpage phishing detection systems. We divided the original dataset into an 80%–20% split for training and testing before running the ML models. The experiments were conducted on an HP Z-book laptop equipped with an Intel® Core™ i7-11800H @ 2.30 GHz processor with 32 GB RAM. To ensure a balanced comparison across the three models, no hyperparameters were tuned for the models, and only the default parameters found in the Sklearn library were used. First, we performed a baseline evaluation, followed by a comparative assessment using three FS methods and previous studies.

4.4.1. Baseline experiment

In the first performance evaluation stage, RF, XGB, and KNN models were trained on the full dataset (87 features). This set up the baseline on which we assessed the actual ability of the FS methods to increase efficiency. The results of this experiment are shown in Table 3.

Table 3
Baseline machine learning results with 87 features.

Model	Accuracy (%)	Precision	Recall	F1-score	Time (s)
RF	96.76	0.97	0.97	0.97	1.651
XGB	96.80	0.97	0.97	0.97	2.294
KNN	83.24	0.83	0.82	0.83	1.212

Note: RF: random forest; XGB: XGBoost; KNN: k-nearest neighbors.

The ensemble models, RF and XGB, outperform KNN in this experiment by a large margin. XGB attains the best accuracy, slightly edging out RF with a 96.80% accuracy, though their difference is a minimal 0.04%. Both models show comparable precision, recall, and F1-scores. However, the significant difference comes in processing time. XGB processes 2,288 samples in 2.294 s, roughly 1 ms per sample, while RF is much quicker at 0.72 ms per sample. Given the minor accuracy variance, RF's rapid response positions it as the preferred selection for phishing detection, which requires swift processing of vast data.

4.4.2. *Determining optimal number of features*

Before delving into the performance evaluation of SLA-FS and the comparison methods, we needed to select the number of top features to be employed by each FS method to ensure a fair comparison. Prior research underscores the importance of establishing a balanced feature set, as excessively reducing or retaining features leads to reduced detection accuracy. Hence, we performed an empirical evaluation using SLA-FS and the best-performing model from the baseline experiment, XGB. We selected four thresholds (top 90%, 75%, 50%, and 25%). The qualifying features from SHAP and LIME rankings were selected for each threshold, and only the features in both readings were selected using Eq. (4). For each threshold experiment, we have (1) 73 features out of 78 (90% threshold), (2) 56 features out of 65 (75%), (3) 34 out of 44 (50%), and (4) 12 out of 22 (25%). The four subsets were then evaluated using XGB, and the results are illustrated in Fig. 4.

Here, it is evident that XGB achieves the highest accuracy (97.21%) when utilizing the top 56 features, which degrades if we increase or decrease the number of selected features. This selection of *m* enables the ML models to bypass less contributory features while maintaining sufficient data for accurate detection. Hence, our main comparative assessment was conducted with each FS method, selecting the top 56 features, which each ML model subsequently utilized.

4.4.3. *Comparative assessment with traditional FS*

In this experiment, we first assessed the performance of SLA-FS alongside information gain (IG), chi-square test order, and RFE. The results are illustrated in Table 4.

The two ensemble models (RF and XGB) significantly outperform

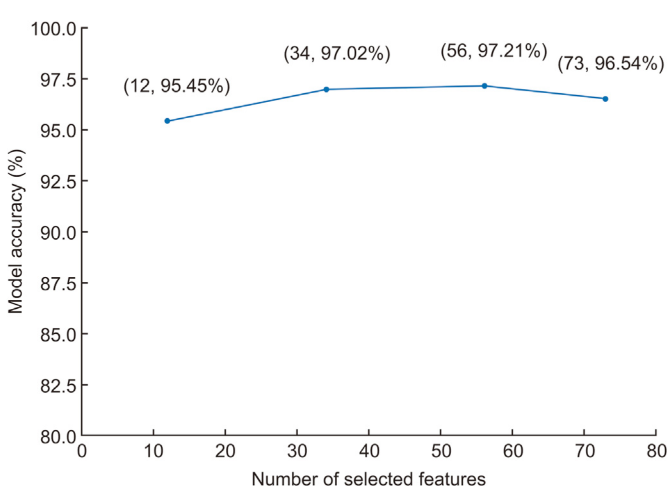

Fig. 4. XGBoost detection accuracy with different feature sets.

Table 4
Comparative feature selection (FS) method evaluation with 56 features.

FS method	Model	Accuracy (%)	Precision	Recall	F1-score	Time (s)
SLA-FS	RF	97.41	0.98	0.97	0.97	0.875
	XGB	97.21	0.97	0.97	0.97	2.596
	KNN	84.51	0.85	0.85	0.84	0.622
IG	RF	96.85	0.97	0.97	0.97	1.149
	XGB	96.89	0.97	0.97	0.97	2.258
	KNN	84.03	0.84	0.84	0.84	0.828
Chi-test	RF	96.63	0.97	0.97	0.97	1.433
	XGB	96.98	0.97	0.97	0.97	2.012
	KNN	83.33	0.83	0.83	0.83	1.057
RFE	RF	97.11	0.97	0.97	0.97	1.911
	XGB	97.02	0.97	0.97	0.97	2.263
	KNN	83.72	0.84	0.83	0.83	1.178

Note: RF: random forest; XGB: XGBoost; KNN: *k*-nearest neighbors; SLA-FS: SHapley Additive exPlanations (SHAP) and local interpretable model-agnostic explanations (LIME)-aggregated feature selection; IG: information gain; RFE: recursive feature elimination.

KNN during the assessment. Specifically, employing SLA-FS, RF achieves an accuracy of 97.41%, narrowly edging out XGB's best of 97.21% (also through SLA-FS). This peak performance, attained while using the combination of SHAP and LIME as the FS, highlights the effect of considering global and local perspectives in web phishing FS. The RFE method comes second, with the top two models dropping their detection ability by 0.3% and 0.19%, respectively. An important finding emerges from the use of wrapper-style FS methods (SLA-FS and RFE) on the dataset: the ensemble models demonstrate enhanced accuracy compared with filter-based FS methods (information gain and chi-test). Additionally, RF emerges superior while employing wrapper-like FS (SLA and RFE) for selection. Still, when applying filter-based methods like information gain and chi-square test, XGB takes the lead, achieving accuracies of 96.89% and 96.98%, respectively. In contrast, KNN's performance significantly trails that of those two despite improvement through FS. The highest this model achieves in terms of accuracy is only 84.51%, using SLA-FS. Furthermore, while RF and XGB maintain close competition in precision, recall, and F1-score around 0.97 across metrics-KNN averages only 0.83 in almost all the assessments, further emphasizing its limitations.

In regard to prediction speed, RF excels, taking only 0.875 s to process the SLA-FS reduced dataset and highlighting its suitability for real-time detection. This performance is notably faster than XGB's 2.596 s attained using our proposed method. Due to the feature set reduction, the RF model processes each sample faster (0.38 ms) than in the first experiment (0.72 ms/sample). The same can be observed for XGB, validating our framework's ability to enhance detection speed. Though the fastest in the second experiment was KNN, its lower accuracy limits its applicability.

By visualizing their confusion matrices, a deeper inspection of the top four combinations of model and FS methods (SLA-FS with RF and XGB and RFE with the same) reveals more insights into our proposed method's applicability. Here, "0" stands for the benign class and "1" stands for the malicious class. First, the lowest number of false positives and false negatives were attained while employing SLA-FS. Though using RF resulted in the least false negatives (where the model predicts a phishing attempt even though the website is safe), it contains 33 false positives. False positives are when a website is classified as benign/normal even though it is a phishing one, and it is the most dangerous error for a detection model. This is the lowest number when XGB is applied after our proposed FS. Model with our proposed SLA-FS method (Fig. 5(a) and (b)) concede less false negatives than the models with RFE FS (Fig. 5(c) and (d)). This further validates the superior ability of our explainable FS method.

4.5. *Comparison with previous FS frameworks*

This section provides a comparative analysis of the impact of existing FS techniques on this dataset and the outcomes of our study. For a

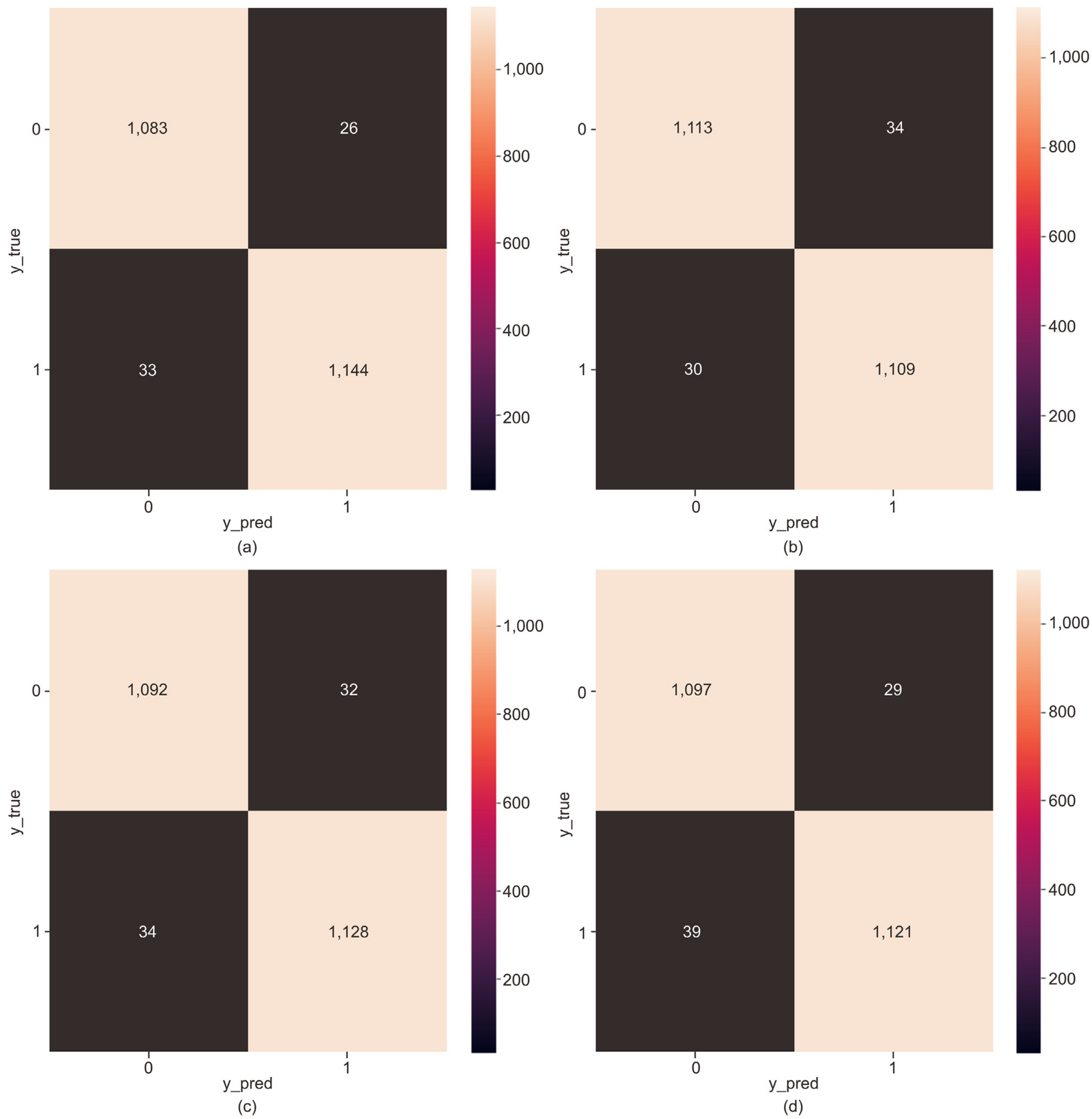

Fig. 5. Comparison of the top two models and top two feature selection (FS) methods. (a) Random forest (RF) with SHAP and LIME-aggregated FS (SLA-FS); (b) XGBoost (XGB) with SLA-FS; (c) RF with recursive feature elimination (RFE); (d) XGB with RFE.

balanced comparison, we specifically focus on applying FS methods to enhance RF and XGB model performance, assessing the accuracy metrics post-FS. Our goal is to distinguish the contributions of proposed FS techniques in boosting model efficacy rather than evaluating the singular efficiency of ML models themselves. Our findings are detailed in Table 5.

Existing works employing RF and XGB have proposed some innovative solutions. Filter and wrapper methods such as chi-square order and univariate FS were applied; however, they only improved RF model's accuracy by 0.22% and 0.04%, respectively. A hybrid method combining power prediction score and RFE was used by Moedjahedy et al. (2022), also to limited success (0.17% increase). While the existing methods enhance phishing attack detection accuracy to some extent, our proposed method outperforms all existing studies that use RF and XGB. The framework enhances phishing detection accuracy of both RF and XGB by

Table 5
Comparative analysis with existing feature selection (FS) methods.

Proposed FS method	ML model	Post-FS accuracy (%)	Improve (%)
Power prediction score + RFE (Moedjahedy et al., 2022)	RF	96.93	0.17
Chi-square order (Hannousse and Yahiouche, 2021a)	RF	96.83	0.22
Univariate FS (Adane et al., 2023)	RF	96.54	0.04
Univariate FS (Adane et al., 2023)	XGB	97.16	0.09
SLA-FS (This study)	RF	97.41	0.65
SLA-FS (This study)	XGB	97.21	0.41

Note: RF: random forest; XGB: XGBoost; SLA-FS: SHapley Additive exPlanations (SHAP) and local interpretable model-agnostic explanations (LIME)-aggregated feature selection; RFE: recursive feature elimination.

0.65% and 0.41%, respectively, which is significantly more than the comparing studies. We conclude that the combined use of SHAP and LIME can thus increase phishing detection accuracy more than many other existing methods, making it a viable tool for further exploration.

5. Discussion and real-world implications of SLA-FS

This study demonstrates the impact of integrating XAI into the FS process for phishing detection, addressing a critical vulnerability in today's Internet landscape. The proposed SLA-FS framework innovatively combines SHAP and LIME to create an integrated set of features, capturing the global view of web traffic data and local variabilities, which derives critical insights for practical applications. This methodology improves the accuracy of phishing detection models over existing methods (Sections 4.4.3 and 4.5) and enhances their interpretability. From a practical standpoint, our method offers significant benefits for managing cybersecurity. The class-specific focus of SLA-FS allows system administrators to tailor FS based on specific attack types or benign traffic, offering tunability and precision, an aspect often ignored in contemporary literature. Security managers can use the explainable approach to focus on important features and gain deeper insights into the decision-making processes of their detection models. Furthermore, the reduced feature set results in faster detection speeds (0.38 ms/sample when using RF), translating to lower computational costs and making our method a cost-effective solution for deployment. This transparency and reliability of the explainable FS approach thus fosters greater trust in AI-based security solutions among stakeholders.

6. Conclusion and future works

The rapid surge in online activity has led to increased webpage phishing attacks. Existing ML detection methods are aided by FS processes, which help the models focus on the important features. However, they often fail to adapt to the constant evolution of webpage phishing attacks, decreasing detection accuracy and lacking a distinct focus on attack types. To combat this threat and enhance practical applicability, this study looks to XAI algorithms for an adaptable, class-specific FS method that assesses a dataset's global representation while considering localized variations to strengthen phishing attack identification. Our SLA-FS framework employs SHAP and an aggregated version of LIME algorithms to produce two sets of feature importance rankings. Features with global and local importance are selected to create a unified set for the ML models. The framework was applied to a recent dataset containing the latest web phishing features, where it selected 56 out of 87 features through an empirical assessment. Experiments showed that the reduced dataset resulted in a 0.65% increase in RF's accuracy, while XGB attained a 0.41% accuracy when compared with the full dataset. This performance was also compared with information gain, chi-square test, and RFE-applied datasets, where the proposed SLA-FS outperformed each, achieving the highest accuracy for RF (97.41%), XGB (97.21%), and KNN (84.51%) while speeding up the detection process. The combination of SLA-FS and RF models also resulted in the highest accuracy obtained across any previous benchmarks on this dataset, validating the use of XAI and the superiority of SLA-FS for webpage phishing FS. While this study solely focused on the FS aspect of phishing detection, our future work will further incorporate ensemble stacking and neural network models to improve attack detection accuracy on top of SLA-FS.

CRediT authorship contribution statement

Sakib Shahriar Shafin: Writing – original draft, Software, Methodology, Data curation, Conceptualization.

Declaration of competing interest

The author declares that there is no conflicts of interest.

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